MACHINE WHITE PAPERS

YCM TV-188 B — Loco Motive

Large-envelope CNC vertical mill for oversized workpieces, complex pocketing, and multi-face machining. 32" W × 70" L × 32" H workpiece envelope.

Loco Motive · large-envelope CNC vertical machining center

Loco Motive is where the oversized work lives — valve bodies, wellhead spool adapters, large manifold blocks, and any part that busts the 20-inch class of mills. YCM's TV-188 platform is a heavier, more rigid class of vertical machining center; the extra column and table mass matters when you're taking heavy roughing cuts in Inconel or hogging out a wellhead block.

What Loco Motive is for

Loco Motive exists for the part that just got quoted somewhere else and came back "exceeds our envelope." 32" × 70" × 32", a 78.7" table, and 4,409 lb of workpiece capacity on a patented T-base — this is a heavier class of vertical machining center than the 20-inch machines most job shops stop at. The BT50 spindle and 25 HP matter as much as the travels: taking a real roughing cut in Inconel on a large block is a rigidity problem before it is a horsepower problem, and mass is what solves it.

Pick this machine when

- The part is prismatic and too big for a 20-inch-class mill — large valve bodies, wellhead spool adapters, manifold blocks.
- The workpiece is heavy enough that table capacity, not travel, is the binding constraint.
- You're hogging a large amount of material out of a solid block and chatter has been the problem elsewhere.

- The part needs multi-face work at size, where re-fixturing on a smaller machine would cost you the tolerance.

> **When it's the wrong machine**
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> Small parts. Putting a 6-inch bracket on this table wastes the machine and costs you money — Pistol Whiskers will do it faster and cheaper. It is also not a high-RPM machine: 6,000 RPM standard is plenty for steel and superalloy at this size, but small-diameter aluminum work that wants 12,000+ RPM is not what this platform is for.

From the floor

The T-base does two jobs — rigidity and chip clearance — and the second one matters more than buyers expect on big Inconel jobs. A large pocketing operation in 718 generates a volume of hot, stringy chip that will re-cut and destroy surface finish if it sits in the pocket. Evacuation strategy gets planned alongside the toolpath, not after it.

Pick your industry

YCM TV-188 B for Oil & Gas

Full spec sheet, typical oil & gas parts, alloys, tolerance expectations, documentation approach.

YCM TV-188 B for Aerospace

Full spec sheet, typical aerospace parts, alloys, tolerance expectations, documentation approach.

YCM TV-188 B for Military & Defense

Full spec sheet, typical defense parts, alloys, tolerance expectations, documentation approach.

YCM TV-188 B for Industrial & General Manufacturing

Full spec sheet, typical industrial parts, alloys, tolerance expectations, documentation approach.

Have a project?